SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I R.YUVASHREE(192521485)(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R.YUVASHREE

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

CO1 - Assessment Tool - 1

CSA1514 - Cloud Computing And Big Data Analytics.

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1. Define cloud computing and Explain four characteristic

Cloud computing is the Delivery of computing services such as servers, storage, databases and software over the internet.

Characteristics

1. On-demand self services: User access resources whenever needed.
2. Broad network access: Services are accessible through the internet.
3. Resource pooling: Resources are shared among multiple users
4. Rapid elasticity: Resources can be increased or decreased quickly
5. Measured service: Users pay based on usage.

Difference: Cloud Computing requires less physical Infrastructure management than traditional on-premise computing.

2. Evolution of Cloud computing

Cloud computing evolved through these Stages.

1. Hardware evolution : Mainframe computers provided centralized computing.
2. Time-Sharing : Multiple users shared computing resources
3. Virtualization : One physical server ran multiple virtual Machines
4. Internet evolution : Internet connectivity enabled remote access.
5. Software evolution : Web application and APIs enabled online software delivery.
6. Modern Cloud : Platforms such as AWS, Azure and Google Cloud provide on-demand and scalable services.

Conclusion : Hardware, Virtualization, networking and Internet software enabled modern cloud computing.

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3. Role of Server Virtualization in Cloud Computing.

Server Virtualization allows multiple Virtual Machines to run on one physical server using a hypervisor

Benifits:

- Improves resource utilization.
- Reduces hardware costs
- Provides isolation between Virtual Machines
- Supports flexible resource allocation.

Difference:

Virtualization	Cloud Computing
Technology that creates VM	Service model that delivers computing resources
Eg: VMware ESXI	Eg : AWS

Conclusion:
Virtualization is an important enabling Technology for cloud computing.

4. Role of Data centers in Cloud Computing.

A data center is a facility containing servers, storage, networking power and cooling systems.

Roles:
1. Provides computing power
2. Stores and manages data.
3. Connects users through networks
4. Supports security and reliability
5. Enables scalability and backup.

Eg: AWS data center support services such as virtual servers and cloud storage.

Conclusion: Data centers are physical foundation of Cloud computing.

5. Differentiate IaaS, PaaS, SaaS and XaaS.

Model	Description	Example
IaaS	Provides virtual servers, storage, and networking	Amazon EC2
PaaS	Provides a platform for developing applications	Google App Engine
SaaS	Provides Ready to use software over the internet	Google Docs
XaaS	Provides various Technology Services as a service	Amazon RDS (DBaaS)

Conclusion:

IaaS - Rent Infrastructure

PaaS - Develop applications

SaaS - Use Software

XaaS - Access Technology services on demand

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